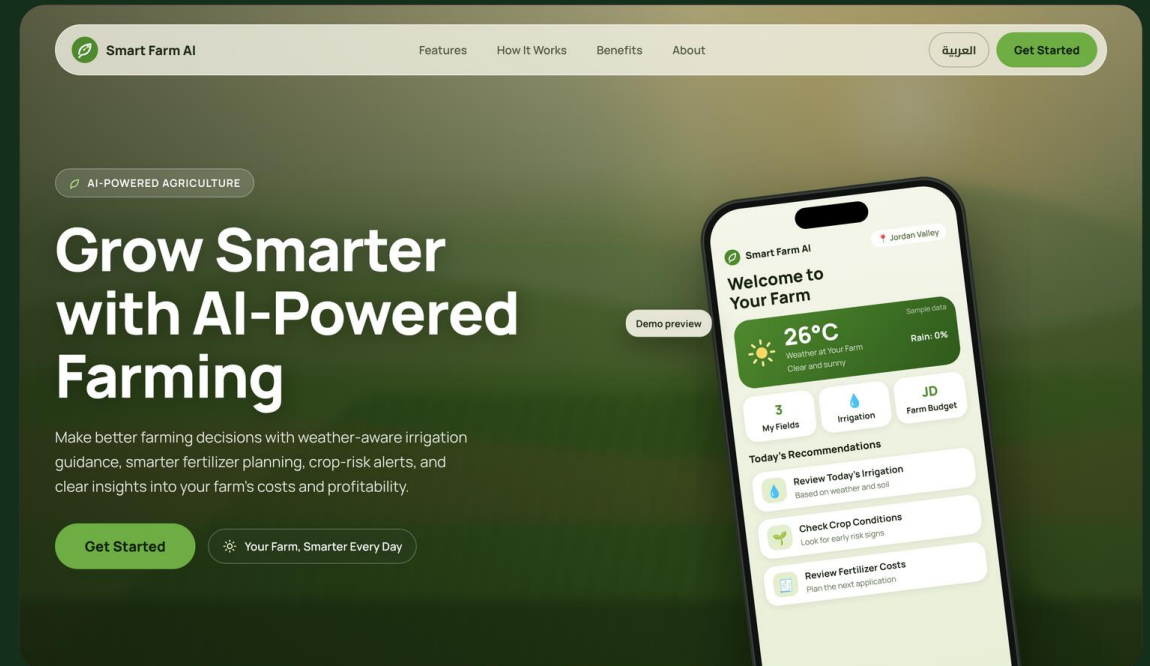


MAS · DECISION SUPPORT FOR TOMATO FARMS

Smart Farm AI

Tomato Farm Assistant

A bilingual Arabic/English application that helps tomato farmers organize farm records, understand costs and revenue, and make better-informed decisions.



Farm information is scattered

WHERE IT LIVES TODAY



Notebooks

Day-to-day notes on irrigation, spending and harvests



Spreadsheets

Costs and sales tracked in separate files



Separate weather services

Forecasts that never meet the farm's own numbers



WHAT THAT MAKES DIFFICULT



Tracking spending

No running total of what the season has cost so far



Telling expected income from actual sales

Estimates and completed sales blur together



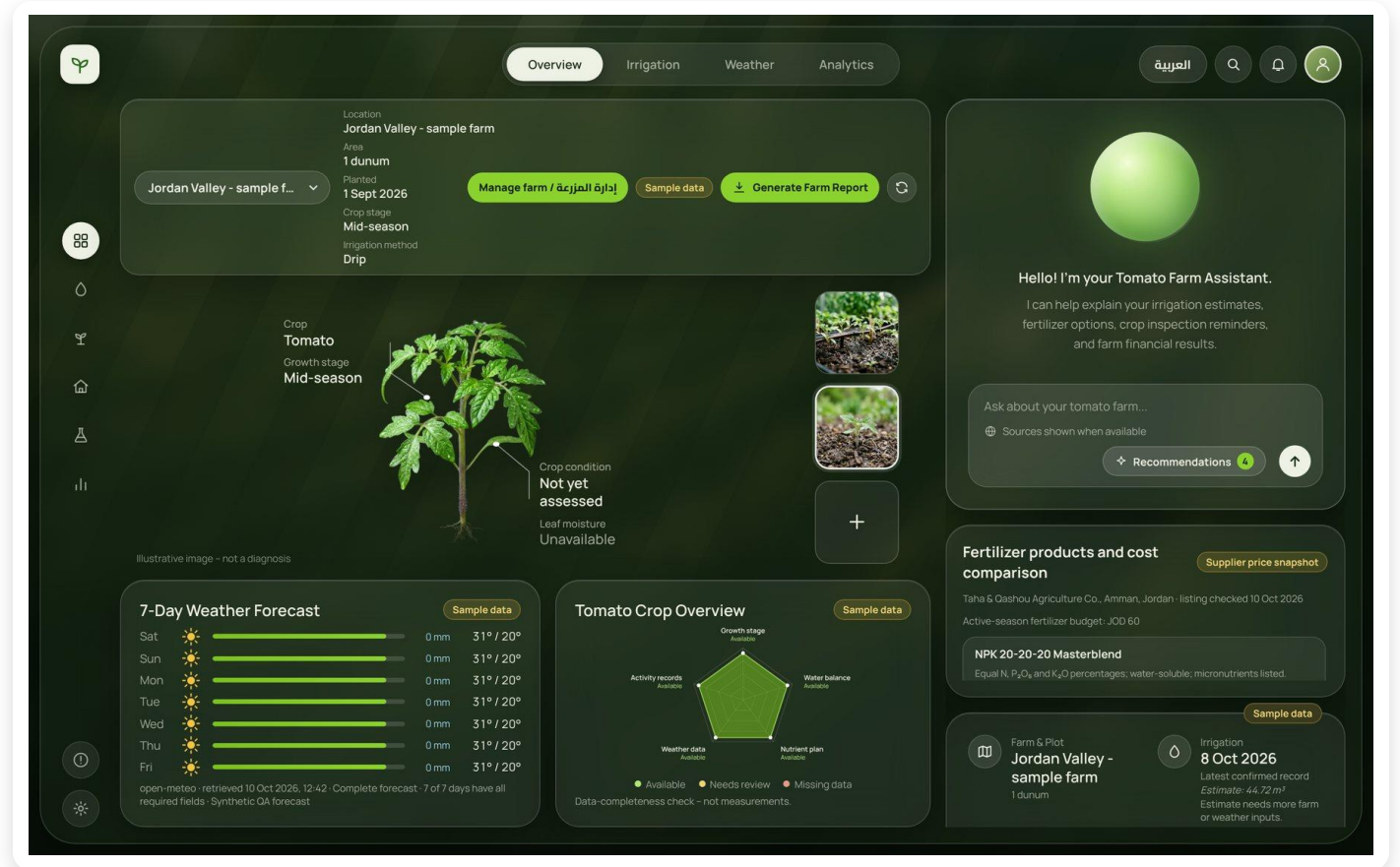
Seeing how conditions change the plan

Less water or a lower price? The effect is a guess

One dashboard for the whole season

Smart Farm AI brings the farm's information into one dashboard, makes its assumptions visible, and explains results in the farmer's preferred language.

- ✓ Records, water and money in one view
- ✓ Assumptions and missing inputs shown beside every estimate
- ✓ Arabic or English at the press of a button



The dashboard running locally with the seeded sample farm and a synthetic test forecast

Five capabilities in one place



Farm management

Records for seasons, irrigation, expenses, harvests and sales



Water planning

Weather-based irrigation estimates that clearly name what is missing



Financial analysis

Actual and projected costs, revenue, profit and break-even price



What-if simulations

Explore changes in water, selling price and costs without changing saved records

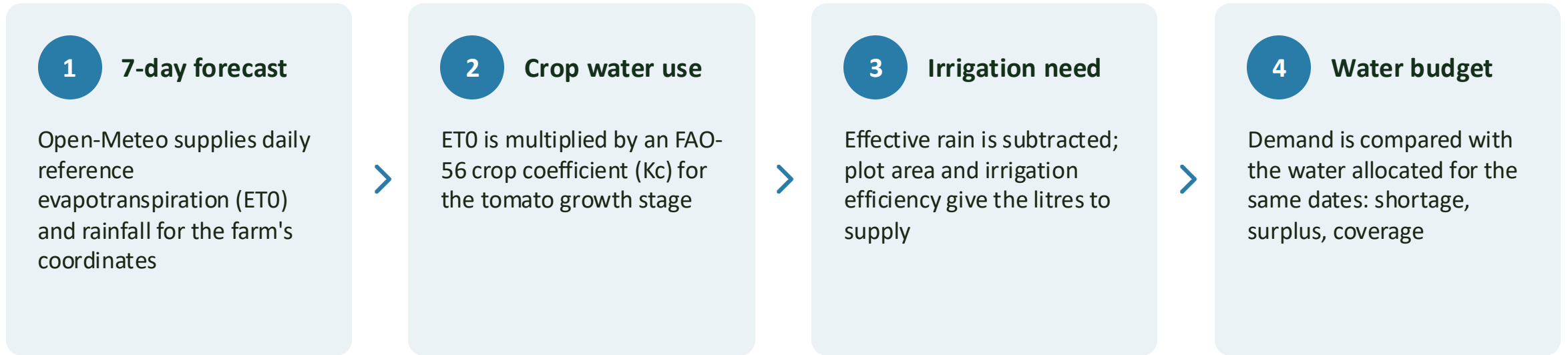


Accessible guidance

A sourced AI assistant, optional Arabic audio and downloadable PDF reports

Supported today: open-field tomato · drip irrigation · amounts in Jordanian dinar (JOD) · Asia/Amman time

Water planning that shows its work



A missing input is reported, never guessed

- No system flow rate: no runtime estimate
- No allocation matching the forecast dates: no shortage figure
- An incomplete forecast: no full-period total

CROP COEFFICIENT (Kc) BY GROWTH STAGE

0.60

Initial

0.875

Development

1.15

Mid-season

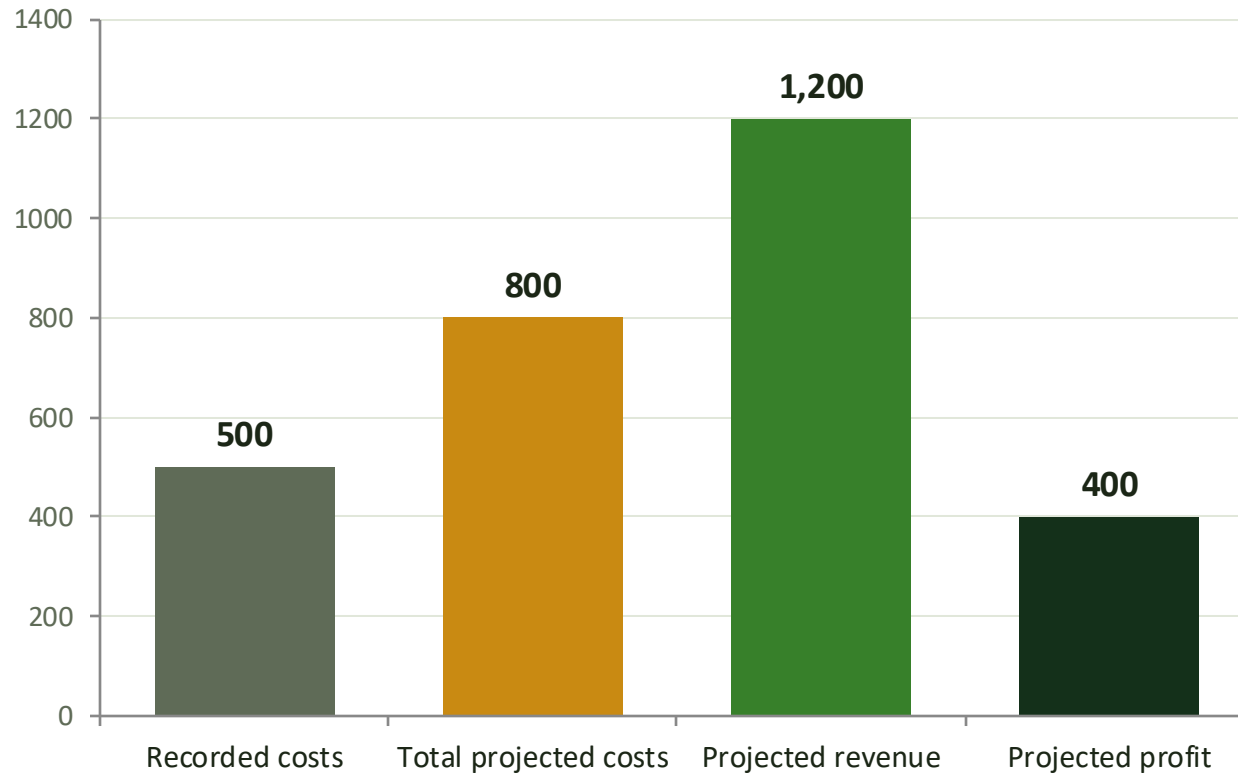
0.80

Late season

FAO-56 reference values. Development is a midpoint approximation; late season is a fixed estimate.

Actual and projected money, kept apart

Worked example from the finance engine (JOD)



Example inputs: 500 JOD already spent, 300 JOD still expected, and 2,000 kg left to sell at an assumed 0.600 JOD/kg.



Actual

Recorded expenses and completed sales only



Projected

Adds the remaining costs and the remaining harvest at an assumed selling price

BREAK-EVEN PRICE IN THE EXAMPLE

0.400 JOD per kg

Total projected cost ÷ (kg sold + kg still to sell) = $800 \div 2,000$.
Below the assumed 0.600, so the example projects a profit.

Simulations never change saved records

CHANGE ANY OF FOUR INPUTS



Available water
litres



Remaining harvest
kg still to sell



Selling price
JOD per kg



Additional costs
JOD

SEE SEVEN RESULTS SIDE BY SIDE

Available water, demand and shortage; total cost, revenue, profit and break-even price. Each shows the baseline, the simulated value and the difference.

- ✓ Nothing is written to the database
- ✓ Baseline and scenario use the same forecast
- ✓ No yield loss is predicted from a water shortage

Simulation comparison													
Not saved. These results do not change farm records or predict yield changes caused by water shortage. Baseline water is a seeded demo value.													
Available water <table><tr><td>Baseline</td><td>4,000 L</td></tr><tr><td>Simulated</td><td>30,000 L</td></tr><tr><td colspan="2">Increase: 26,000 L</td></tr></table>	Baseline	4,000 L	Simulated	30,000 L	Increase: 26,000 L		Estimated water demand <table><tr><td>Baseline</td><td>44,722.222 L</td></tr><tr><td>Simulated</td><td>44,722.222 L</td></tr><tr><td colspan="2">No change</td></tr></table>	Baseline	44,722.222 L	Simulated	44,722.222 L	No change	
Baseline	4,000 L												
Simulated	30,000 L												
Increase: 26,000 L													
Baseline	44,722.222 L												
Simulated	44,722.222 L												
No change													
Estimated shortage <table><tr><td>Baseline</td><td>40,722.222 L</td></tr><tr><td>Simulated</td><td>14,722.222 L</td></tr><tr><td colspan="2">Decrease: 26,000 L</td></tr></table>	Baseline	40,722.222 L	Simulated	14,722.222 L	Decrease: 26,000 L		Total projected costs <table><tr><td>Baseline</td><td>520.5 JOD</td></tr><tr><td>Simulated</td><td>595.5 JOD</td></tr><tr><td colspan="2">Increase: 75 JOD</td></tr></table>	Baseline	520.5 JOD	Simulated	595.5 JOD	Increase: 75 JOD	
Baseline	40,722.222 L												
Simulated	14,722.222 L												
Decrease: 26,000 L													
Baseline	520.5 JOD												
Simulated	595.5 JOD												
Increase: 75 JOD													
Projected revenue <table><tr><td>Baseline</td><td>1,365 JOD</td></tr><tr><td>Simulated</td><td>1,065 JOD</td></tr><tr><td colspan="2">Decrease: 300 JOD</td></tr></table>	Baseline	1,365 JOD	Simulated	1,065 JOD	Decrease: 300 JOD		Projected profit <table><tr><td>Baseline</td><td>844.5 JOD</td></tr><tr><td>Simulated</td><td>469.5 JOD</td></tr><tr><td colspan="2">Decrease: 375 JOD</td></tr></table>	Baseline	844.5 JOD	Simulated	469.5 JOD	Decrease: 375 JOD	
Baseline	1,365 JOD												
Simulated	1,065 JOD												
Decrease: 300 JOD													
Baseline	844.5 JOD												
Simulated	469.5 JOD												
Decrease: 375 JOD													
Break-even price <table><tr><td>Baseline</td><td>0.2263 JOD/kg</td></tr><tr><td>Simulated</td><td>0.2589 JOD/kg</td></tr><tr><td colspan="2">Increase: 0.0326 JOD/kg</td></tr></table>	Baseline	0.2263 JOD/kg	Simulated	0.2589 JOD/kg	Increase: 0.0326 JOD/kg								
Baseline	0.2263 JOD/kg												
Simulated	0.2589 JOD/kg												
Increase: 0.0326 JOD/kg													
Both columns use the same forecast. Unavailable values require more inputs; this comparison is advisory.													

Comparison cards from a local run on the sample farm

An assistant that cannot invent numbers

1

Reviewed passages are retrieved

The question, in Arabic or English, is matched by keyword against a small curated knowledge file

2

The language model only selects

Groq returns a topic and a source ID as JSON; its own wording and numbers are never copied into the answer

3

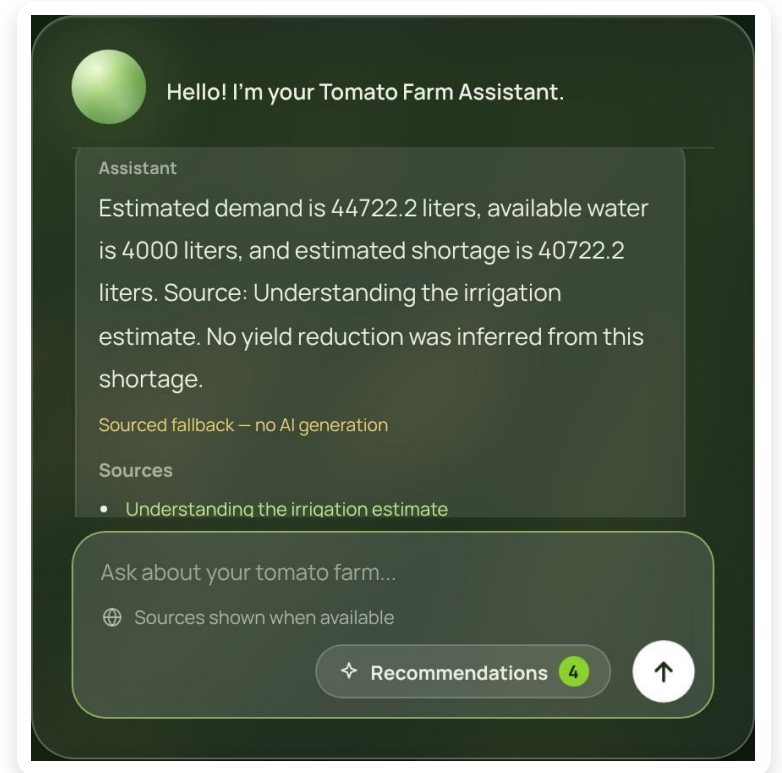
The server writes the answer

Fixed bilingual templates are filled with the farm's own calculator results, units included

4

Sources and limits come attached

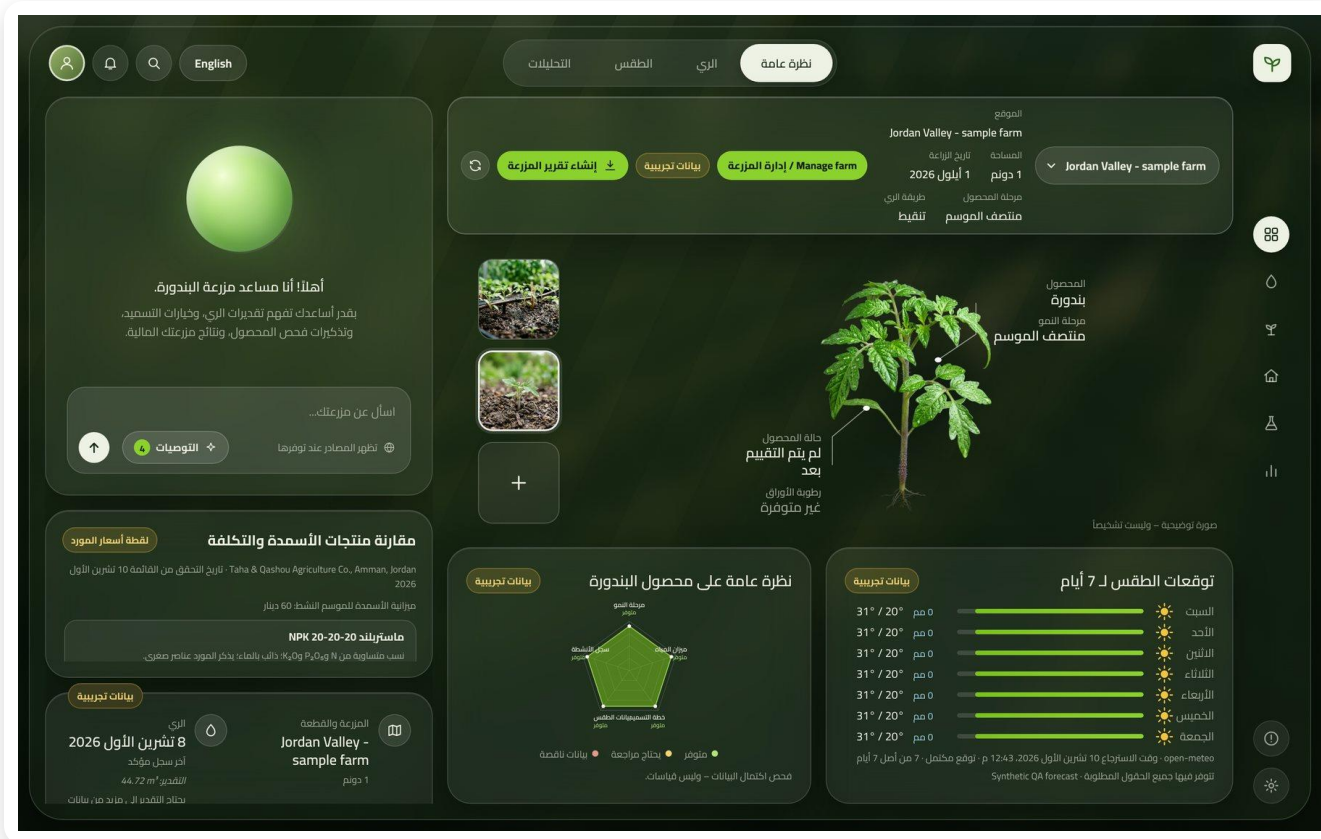
Replies cite the passage used, or say that none was found, and state their limits. If the model fails, a deterministic fallback answers



Reply to “Is the available water enough for this week?” in a local run without a model key: the fallback path, labelled as such.

Knowledge sources: FAO-56, UC IPM, Penn State Extension

Guidance in the farmer's language



The same dashboard in Arabic, laid out right to left



Arabic and English throughout

Interface, recommendations and assistant answers, with a full right-to-left layout



Recommendations read aloud

Optional Jordanian Arabic voice from Azure AI Speech; keys stay on the server



Downloadable PDF reports

A farm report in either language, with correctly shaped Arabic text



Advice that explains itself

Each card gives an action, the reason, the evidence and its limits

How the system is built

Web dashboard	HTML, CSS and plain JavaScript, with PDF reports generated in the browser
FastAPI	REST endpoints for farms, seasons, records, simulation, assistant and voice
Services	Dashboard builder, recommendations, assistant with retrieval, simulation
Calculation engines	Deterministic Python: irrigation (FAO-56), water budget and finance
Database	SQLAlchemy with Alembic migrations; SQLite locally or PostgreSQL 16 in Docker

OPTIONAL EXTERNAL SERVICES



Open-Meteo

7-day weather forecast, cached per farm



Groq

Language model for the assistant



Azure AI Speech

Arabic text-to-speech

The app keeps working without any of them: a missing forecast shows as unavailable and the assistant falls back to a deterministic answer.

Built to be trusted, and tested

778

Python tests passing

17

JavaScript front-end tests passing

8

PostgreSQL concurrency cases reported in the handoff notes

Python and JavaScript suites re-run for this deck on 10 October 2026.



Unknown is never zero

A missing value stays empty and comes with the reason it is missing



Every figure names its origin

Live, cached, sample and simulated data are labelled as such



Retries are safe

A repeated save returns the original result; a conflicting one is refused



Money is exact

Decimal arithmetic, dinars to three places, never silently rounded

What it is, and what it is not



A decision-support demonstration



Shows records, water needs and finances together



Covers open-field tomato with drip irrigation



Runs locally for a single demo owner



Advisory only: recommendations never write farm records



Not a diagnosis or a guarantee



Does not diagnose plant disease or prescribe treatment



Does not guarantee yield or profit



Does not predict crop loss from a water shortage



Not ready for public use: no sign-in or multi-user separation yet

One dashboard, visible assumptions, the farmer's language



Records, water and money in one place



Estimates that state what they assume and what is missing



Arabic and English guidance, with sources

RUN IT LOCALLY

```
1  copy .env.example to .env
2  pip install -r requirements.txt
3  alembic upgrade head
4  python -m backend.app.seed
5  uvicorn backend.app.main:app --port 8001
```

Then open <http://127.0.0.1:8001/dashboard.html>